pfi

Python for Fixed Income

COURSE 9 · DEPLOYMENT

Deployment and Your Accounts

WEEK 1 · DAY 1

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: deployment and your accounts

Today

1. What deployment is, and where each piece runs
2. The free route, as each platform reads today
3. Your accounts: GitHub, and Hugging Face today
4. What origin is
5. A remote you own: push, clone, fetch, pull
6. Before anything goes public: public_scan
7. The first push to GitHub, masked
8. Masking a log before you share it

Every cloud step, two ways

On your own account

the PowerShell you run

A recorded transcript

the course's run, masked

No notebook cell reaches a platform.

Today has no server: the map, the accounts, a remote you own, and the scan.

The rules for deployed code

Checked first	nothing goes public until public_scan is empty
Synthetic bonds	a real company only as the author of a document
Described, never sold	a price or DV01 per 100 of face, with its as-of date
Firm policy first	free tiers are personal accounts for learning

On a platform's figures

Allowed names

included in Hobby as read

the free tier as read

the platform line

Never called

guaranteed

unlimited

free forever

What deployment is

Laptop

write and test

→

Repository

others read it

→

Test machine

nobody owns it

→

Server

a URL for anyone

Week 1	bondmath and its tests: a public repo; tests on Linux
Week 2	a bond service at a public URL; the desk's brief from saved runs
Week 3	one image in Docker; logs, drift, secrets, a cost meter

Every step after the first puts your code on a machine you do not sit at.

The free route, as read today

GitHub

public repos; Actions on public repos

Vercel Hobby

1,000,000 invocations, 4 CPU hours

Hugging Face

2 ZeroGPU Spaces after 30 days

Docker Personal

$0, for individual developers

GitHub Actions: free on public repositories (docs.github.com/en/billing/concepts/product-billing/github-actions, read 2026-10-12).

Vercel Hobby: 1,000,000 invocations and 4 CPU hours a month included (vercel.com/docs/plans/hobby, read 2026-10-12).

Hugging Face: 2 free ZeroGPU Spaces after 30 days, PRO $9 a month (huggingface.co/docs/hub/spaces-zerogpu and huggingface.co/pricing, read 2026-10-12).

Docker Personal: $0 (docker.com/pricing, read 2026-10-12).

The paid route is named where it fits, and never required: no lesson asks for a card.

Your accounts

GitHub

github.com/signup; verify your email

Settings, Emails:

Keep my email addresses private

commits then carry your noreply address

Hugging Face, today

huggingface.co/join; confirm your email

Settings, Access Tokens, New token

role: Write

`my_bondmath\.env:  HF_TOKEN=`

2026-10-12

the account is created

2026-10-21

day 10 at one notebook a day

2026-11-12

qualifies: older than 30 days

```
HF token loaded: False.
```

What origin is

What the platform printed, masked

```
Recorded 2026-10-12 on the course's own account; masked
> git remote -v
origin  https://github.com/student/python-fixed-income.git (fetch)
origin  https://github.com/student/python-fixed-income.git (push)
```

A remote: a name for another copy of the same repository.

origin: the one you cloned from. You can read from it; it is not yours to push to.

A remote you own

Work folder

your commits

→

remote.git

bare: history only

→

colleague

a clone

```
branch 'main' set up to track 'origin/main'.
To remote.git
 * [new branch]      main -> main
```

```
## main...origin/main [behind 1]
arrived in origin/main, not yet in main:
1a61175 Ignore the colleague's clone
```

`push`	sends your commits to the remote
`fetch`	copies them to origin/main; main stays
`pull --ff-only`	moves main forward, or refuses

Before anything goes public

env_file

.env at any depth

key_like

sk-, hf_, ghp_, ...

email

any but the course's

large_file

over max_mb

```
file  line     kind                                    detail
     .env     0 env_file        an environment file; never published
     .env     1 key_like sk-or- pattern, starts 'sk-', 18 characters
notes.txt     1    email     an address, starts 'des', 28 characters
problems: 3; safe to publish: no
```

problems: 0; safe to publish: yes; the tests in the folder alone: 40 passed

.gitignore protects a commit; the scan protects the folder. It never prints what it found.

The first push to GitHub

```
git init -b main
git config user.name "<your name>"
git config user.email "<noreply>"
git add .
git commit -m "..."
git remote add origin <your repo>
git push -u origin main
```

What the platform printed, masked

Recorded 2026-10-12, the course's account, masked

```
> git push -u origin main
branch 'main' set up to track 'origin/main'.
To https://github.com/student/pfi-deploy-demo.git
 * [new branch]      main -> main
```

The same lines as the push to remote.git. Git Credential Manager signs you in: no token typed.

Masking a log before you share it

```
To https://github.com/student/pfi-bondmath.git
Connected from <ip>
Production: https://student.vercel.app (<id>)
```

account name	`student`
IPv4 address	`<ip>`
*.vercel.app, *.hf.space	`student`
dpl_, prj_, team_ ids	`<id>`

Longer names go first, so a team slug goes whole. Read the result before you share it.

Excel and Git

Excel side by side: none today

Excel has no remote, no clone, no push, and no scan of a folder. Week 3 brings it back.

```
deploykit day 1: run_git, public_scan, mask_transcript, show_transcript, 6 tests
Ignore the colleague's clone
Start deploykit beside bondmath and its tests: run_git, 1 test
```

```
.gitignore:3:.env   .env
```

Three commits: on the work folder, on remote.git, and in the colleague's clone. 6 tests.

The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: two globs at the top of the folder: deploy/config/.env.production with a key scans clean.

Code that runs is not the same as code that is right.

What you can do now

- Say what deployment is: four places
- Read the free route, with the page and the date
- Open GitHub and Hugging Face; the token in .env
- A remote you own: push, clone, fetch, pull
- public_scan before anything goes public; mask logs

Week 1, Day 2 Branches and pull requests